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```
BEGIN
  //Input seating Plan
  PRINT "Enter number of rows: "
  INPUT rows
  PRINT "Enter number of seats per row: "
  INPUT cols

  DECLARE matrix [rows] [cols]
  FOR i=0 TO rows-1
    FOR j=0 TO cols-1
        INPUT matrix [i] [j]
    NEXT j
  NEXT i

  //Display Seating Arrangement
  PRINT column headers (0 to cols-1)
  PRINT "   " //leading spaces to allign with row labels
  FOR j=0 TO cols-1
    PRINT j + " "
  NEXT j
  PRINT ("/n") //New line.

  FOR i=0 TO rows-1
    PRINT "R" + i + " "
    FOR j=0 TO cols-1
        IF matrix [i] [j] == 0 THEN
            PRINT " _ "
```


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        ELSE
            PRINT " x "
        ENDIF
    NEXT j
    PRINT "/n"
NEXT i

// 3. Seat availability Report
total_free = 0
PRINT "Free seats per row:"
FOR i = 0 TO rows - 1
    row_free = 0
    FOR j = 0 TO cols - 1
        IF matrix[i][j] == 0 THEN
            row_free = row_free + 1
        ENDIF
    NEXT j
    PRINT row_free + " "
    total_free = total_free + row_free
NEXT i
PRINT " Total Free: " + total_free
PRINT "/n"

// Group Booking & Validation.
PRINT "Enter number of customers (K):"
INPUT K
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WHILE (k <= 0 OR k > cols) DO
    PRINT "Invalid. Enter valid group size:"
    INPUT k
ENDWHILE

// 5. Seat Allocation Rules (Manual Middle-Out Search)
mid = rows/2
found = FALSE
dist = 0.

WHILE (dist <= rows AND found == FALSE) DO
    //Check upper row (smaller row index first).
    IF (mid - dist) >= 0 THEN
        r = mid - dist
        consecutive = 0
        FOR c = 0 TO cols - 1
            IF matrix[r][c] == 0 THEN
                consecutive = consecutive + 1
                IF consecutive == k THEN
                    found = TRUE
                    booked_row = r
                    start_seat = c - k + 1
                    BREAK
                ENDIF
            ELSE
                consecutive = 0
            ENDIF
        NEXT c
```

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    ENDIF

    // Ched lower row if no match found or
       it isn't mid row being double checked.
    IF((found == FALSE AND dist > 0) AND(mid+dist)
       < rows)) THEN
       r = mid + dist
       consecutive = 0
       FOR c = 0 TO cols - 1
          IF matrix[r][c] == 0 THEN
             consecutive = consecutive + 1
             IF consecutive == k THEN
                found = TRUE
                booked_row = r
                start_seat = c - k + 1
                BREAK
             ENDIF
          ELSE
             consecutive = 0
          ENDIF
       NEXT c
    ENDIF

    dist = dist + 1
ENDWHILE.
```

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// Confirm Booking & Handle Unsuccessful.
IF found == TRUE THEN
    // Mark seats as booked.
    FOR c = start_seat TO start_seat + k - 1
        matrix[booked_row][c] = 1
    NEXT c

    PRINT "Seated in Row", booked_row,
        "Seats", start_seat, "to",
        startseat + k - 1.
    PRINT "$\n"

    // Display Updated seating Arrangement
     (Explicit Loop)
    PRINT "   "
    FOR j = 0 TO Cols - 1
        PRINT j + " "
    NEXT j
    PRINT "$\n"

    FOR i = 0 TO rows - 1
        PRINT "R" + i + " "
        FOR j = 0 TO cols - 1
            IF matrix[i][j] == 0 THEN
                PRINT "_ "
            ELSE
                PRINT "X"
    END IF
```

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